PRE-MEDICAL : ENTHUSIAST COURSE PHASE -

ENTHUSIAST, LEADER & ACHIEVER COURSE

PHASE : ALL ENTHUSIAST, MLA, MAZA, MAZB, MAZC, MAZD, MAZL, MAZN, MAZO, MAAX, MAAY, MAPA, MAPB, LAKSHYA

TARGET : PRE MEDICAL 2025

Test Type : **MAJOR** Test Pattern : **NEET (UG)**

TEST DATE : 29-03-2025

ANSWER KEY

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
A.	2	4	3	4	3	1	3	1	2	3	3	1	1	3	3	3	2	3	1	3	4	1	3	3	1	3	2	2	4	4
Q.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
A.	1	3	2	4	2	4	2	2	4	1	3	4	2	2	1	3	3	1	1	4	4	3	4	4	4	3	3	2	3	3
Q.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
A.	1	3	2	2	2	4	4	3	3	4	4	3	2	3	3	2	2	3	3	3	4	3	3	4	3	3	2	3	3	2
Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
A.	4	2	4	3	4	3	3	4	3	3	4	4	3	4	2	2	3	4	1	4	3	2	1	4	2	3	3	3	2	4
Q.	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	2	1	1	2	2	3	2	4	2	2	1	3	4	4	4	1	2	3	4	2	1	3	2	4	3	1	4	1	4	4
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170	171	172	173	174	175	176	177	178	179	180
A.	3	4	1	4	2	3	2	2	2	1	2	2	4	4	3	3	2	2	4	4	1	1	3	1	2	4	1	3	4	3

HINT – SHEET

1. **Ans (2)**
XI NCERT Page. No :- 137

2. **Ans (4)**
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3. **Ans (3)**
XI NCERT Page No. # 138

4. **Ans (4)**
NCERT, Pg. # 138, 139

5. **Ans (3)**
NCERT-XI, Pg # 138

6. **Ans (1)**
XI NCERT Pg # 145

7. **Ans (3)**
NCERT XI Pg. # 146, 147
Bundle sheath cells in C_4 plants contain RuBisCo enzyme.

8. **Ans (1)**
NCERT-XI, Pg. # 156

9. **Ans (2)**
NCERT, Pg. # 156, Diagram

10. **Ans (3)**
NCERT Pg. No. # 160

11. **Ans (3)**
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12. **Ans (1)**
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13. **Ans (1)**
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14. **Ans (3)**
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15. **Ans (3)**
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16. **Ans (3)**
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17. **Ans (2)**
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19. **Ans (1)**
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20. **Ans (3)**
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21. **Ans (4)**
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22. **Ans (1)**
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25. **Ans (1)**
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28. **Ans (2)**
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29. **Ans (4)**
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30. **Ans (4)**
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32. **Ans (3)**
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Cortex, Pericycle, Pith

33. **Ans (2)**
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35. **Ans (2)**
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36. **Ans (4)**
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48. **Ans (1)**
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49. **Ans (1)**
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51. **Ans (4)**
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68. Ans (3)
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87. Ans (2)
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88. Ans (3)
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89. Ans (3)
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90. Ans (2)
NCERT Pg # 62

91. Ans (4)
OA ⟶ Pressure constant ⟶ Isobaric Process

OD ⟶ Volume constant ⟶ Isochoric Process

OB ⟶ Isothermal Process

OC ⟶ Adiabatic Process

92. Ans (2)

$$\eta = 1 - \frac{T_2}{T_1} \quad \left[\begin{array}{l} T_1 \rightarrow \text{Temperature of source} \\ T_2 \rightarrow \text{Temperature of sink} \end{array}\right.$$

$$0.4 = 1 - \frac{T_2}{500} \Rightarrow T_2 = 0.6 \times 500 = 300\,K$$

Now :- Temperature of source is T_1'

$$\eta = 1 - \frac{T_2}{T_1'} \Rightarrow 0.5 = 1 - \frac{300}{T_1'}$$

$\therefore T_1' = 600$ K

93. Ans (4)
For a diatomic gas :-

$C_P = 7/2\ R$, $C_V = 5/2\ R$

$$\therefore \frac{C_P}{C_V} = \frac{7R/2}{5R/2} = \frac{7}{5}$$

94. Ans (3)
$Q = \Delta U + W$
$\because \Delta U$ is same

$W_i = +Ve$
$W_{ii} = 0$
$W_{iii} = -Ve$
$\therefore Q_i > Q_{ii}$

95. Ans (4)

$$\lambda = \frac{1}{\sqrt{2}\pi d^2 N}$$

$$\lambda \propto \frac{1}{d^2 N}$$

96. Ans (3)
Isochoric ⟶ V = constant $\Rightarrow \boxed{W = 0}$

Isothermal ⟶ T = constant $\Rightarrow \boxed{\Delta U = 0}$;

Adiabatic $\Rightarrow \Delta Q = 0$

Isobaric ⟶ P = constant

97. Ans (3)

$$\frac{T_F - 32}{180} = \frac{T_K - 273}{100}$$

$$\frac{T - 32}{180} = \frac{T - 273}{100}$$

$5T - 160 = 9T - 2457 \Rightarrow 4T = 2297$

$T = 574.25$

98. Ans (4)
$PV = nRT$

$$PV = \frac{N}{N_A}RT$$

$$PV = NKT \Rightarrow N = \frac{PV}{KT}$$

$$\therefore \frac{N_1}{N_2} = 4 : 1$$

99. Ans (3)
$\because$ Density of water is maximum at 4ºC.

100. Ans (3)

Total energy : $E = \frac{3}{2}PV$

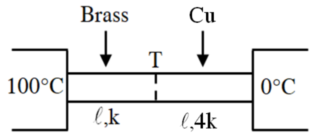
For H_2 : $1500 = \frac{3}{2}PV$ ....(1)

For N_2 : $E' = \frac{5}{2}(2P)V$ ....(2)

$$\therefore \frac{E'}{1500} = \frac{10}{3}$$

E' = 5000 J

101. Ans (4)



$$\because \frac{\Delta Q}{\Delta t} = \text{constant}$$

$$\therefore \frac{kA(100 - T)}{\ell} = \frac{4kA(T - 0)}{\ell}$$

$100 - T = 4T$

$T = 20°C$

102. Ans (4)

By NLC

In first, 20 min

$T_1 = 85°C$, $T_2 = 85 - 15 = 70°C$

then $\frac{85-70}{20} = k\left(\frac{85+70}{2} - 55\right)$ ....(i)

After next 20 min temp. of body will be T_3.

$\frac{70-T_3}{20} = k\left(\frac{70+T_3}{2} - 55\right)$ .... (ii)

By eq. (i) & (ii)

$T_3 = 62.5°C$

103. Ans (3)

$\frac{dQ}{dt} = e\sigma AT^4$

$\frac{(dQ/dt)_1}{(dQ/dt)_2} = \frac{T_1^4}{T_2^4}$

$\frac{10}{(dQ/dt)_2} = \left(\frac{500}{1000}\right)^4 = \frac{1}{(2)^4}$

$(dQ/dt)_2 = 160$ cal.

104. Ans (4)

$\frac{dQ}{dt} \propto AT^4$

$\frac{dQ}{dt} \propto r^2T^4$

$\frac{(dQ/dt)_1}{(dQ/dt)_2} = \left(\frac{1}{2}\right)^2\left(\frac{2\times10^3}{1\times10^3}\right)^4$

$= 4/1$

105. Ans (2)

$kx = mg \Rightarrow k = \frac{mg}{x}$

$T = 2\pi\sqrt{\frac{M+m}{k}} = 2\pi\sqrt{\frac{(M+m)x}{mg}}$

106. Ans (2)

$5T = T + \frac{T}{4}$

it means when bigger pendulum makes one full oscillation the smaller pendulum will make $\left(1+\frac{1}{4}\right)$ oscillation. Thus phase difference = 90°.

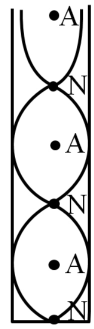
107. Ans (3)

time taken $= \frac{T}{6} = \frac{4}{6} = \frac{2}{3}$sec

108. Ans (4)

$V_{He} = 460 \times \sqrt{\frac{25}{21} \times 8} \simeq 1420$ m/s

109. Ans (1)



In second overtone :-

Nodes :- 3

Antinodes :- 3

Third overtone frequency of closed pipe means seventh harmonic. Which is 7 times the fundamental frequency.

110. Ans (4)

$k \propto \frac{1}{\ell}$ if length is halved k will become two times

$\because T = 2\pi\sqrt{\frac{m}{k}}$

so, time period will become half for the given conditions.

111. Ans (3)

$\Delta x = 6.4 - 6.0 = 0.4$ m

For minimum intensity $\Delta x = (2N+1)\,\lambda/2$

$= \frac{(2N+1)}{2} \times \frac{V}{n}$

$\therefore n = \frac{(2N+1)V}{2\Delta x} = \frac{(2N+1)320}{2\times0.4} = 400\,(2N+1)$ Hz

If N = 0 then n = 400 Hz, if N = 1 then n = 1200 Hz

If N = 2 then n = 2000 Hz, if N = 3 then n = 2800 Hz

112. Ans (2)

Fundamental frequency of pipe

$n_0 = \frac{V}{4\ell} = \frac{330}{4\times0.15}$

$n_0 = 550$ Hz

The available overtones are $3n_0$, $5n_0$, $7n_0$, $9n_0$

So overtone frequencies are

1650, 2750, 3850, ... etc.

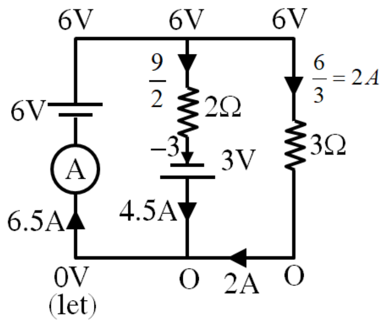
113. Ans (1)

Potential difference across 2 ohm.

Resistance is 24V

So, $I = \frac{V}{R} = \frac{24}{2} = 12A.$

114. Ans (4)



115. Ans (2)

Electric field inside a current carrying conductor is due to potential difference created by battery.

116. Ans (3)

$J = \sigma E$

$E \propto J \propto \frac{1}{A}$

So, $E_Q > E_P$

117. Ans (3)

$R_{AB} = \frac{8}{6} = \frac{4}{3}\Omega$

$R_{CD} = \frac{3}{5}\Omega$

$\frac{R_{AB}}{R_{CD}} = \frac{4/3}{3/5} = \frac{20}{9}$

118. Ans (3)

$100 = \frac{(330)^2}{R} = \frac{(220)^2}{R'}$

$R' = R\left(\frac{220}{330}\right)^2 = R\left(\frac{4}{9}\right)$

$= \frac{4R}{9}$

119. Ans (2)

$F = \frac{1}{4\pi\epsilon_0}\frac{q_1q_2}{r^2}$

120. Ans (4)

$n = \frac{q}{e} = \frac{2 \times 10^{-6}}{1.6 \times 10^{-19}} = 1.25 \times 10^{13}$

$(n_{total})_{electron} = 2.25 \times 10^{13} + 1.25 \times 10^{13}$

$= 3.50 \times 10^{13}$

121. Ans (2)

Fact

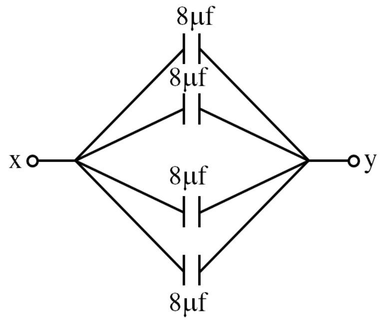
122. Ans (1)

$w = \vec{F} \cdot \overrightarrow{\Delta r} = q_0\vec{E} \cdot \overrightarrow{\Delta r}$

$= q_0 ((E_x\hat{i} + E_y\hat{j} + E_z\hat{k}) \cdot (a\hat{i} + b\hat{j})$

$= q_0 (E_xa + E_yb)$

123. Ans (1)



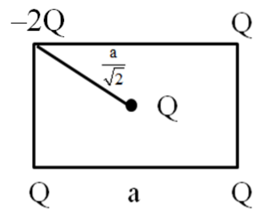
$C_{xy} = 32\ \mu F$

124. Ans (2)

(A), (B), (D) are correct statements.

(C) is incorrect because charge of a proton is equal to sum of charge of 2 up quark and 1 down quark but it might seem reasonable that simply adding up the quarks' masses would give you a protons mass ; yet it doesn't.

125. Ans (2)



$E = \frac{k(3Q)}{\left(\frac{a}{\sqrt{2}}\right)^2}$

$= \frac{6kQ^2}{a^2}$

126. Ans (3)

$V = \frac{KQ}{R} - \frac{Kq}{x}$

127. Ans (2)

$w = q\Delta v$

$= 1\left[\frac{9 \times 10^9 \times 20 \times 10^{-6}}{25} + \frac{9 \times 10^9 \times 10 \times 10^{-6}}{25}\right]$

$= \frac{9 \times 10^3}{25}\ [30]$

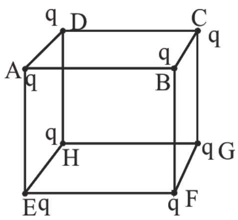
$= 10.8 \times 10^3 J$

128. Ans (4)

$U = U_1 + U_2 = -pE\cos\theta + \frac{k(q)(-q)}{2\ell}$

$= -pE\cos\theta - \frac{kq^2}{2\ell}$

129. Ans (2)



By symmetry ϕ due to charge at E, F, G & H from face ABCD are equal

ϕ due to charge at $E = \underbrace{\left(\frac{q}{8}\right)}_{q_{in}} \times \frac{1}{\epsilon_0} \times \underbrace{\frac{1}{3}}_{3\text{ face}}$

ϕ due to charge at E, F, G, H $= 4 \times \frac{q}{24\epsilon_0} = \frac{q}{6\epsilon_0}$

ϕ due to charge at A, B, C & D = 0

field lines are perpendicular to area vector

130. Ans (2)

Potential equal $= \frac{KQ_a}{a} = \frac{KQ_b}{b}$

$\frac{E_a}{E_b} = \frac{\frac{KQ_a}{a^2}}{\frac{KQ_b}{b^2}} = \frac{b}{a}$

131. Ans (1)

$F = (Q)\left(\frac{V}{2d}\right)$

$= (CV)\left(\frac{V}{2d}\right)$

$F = \frac{CV^2}{2d}$

132. Ans (3)

$C_m = KC_{air}$

$E_m = \frac{E_{air}}{K}$

$V_m = \frac{V_{air}}{K}$

133. Ans (4)

$U = \frac{1}{2}C\left(V_2^2 - V_1^2\right)$

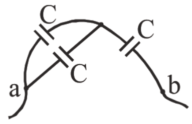
$U = \frac{1}{2} \times 6 \times 10^{-6}(400 - 100)$

$= 900\ \mu J$

134. Ans (4)

At t = 0 whole current passes through capacitance; so effective resistance of circuit is 1000Ω and current .i $= \frac{2}{1000} = 2 \times 10^{-3}$A = 2mA. After sufficient time, steady state is reached; then there is no current in capacitor branch; so effective resistance of circuit is 1000 + 1000 = 2000Ω and current i $= \frac{2}{1000} = 1 \times 10^{-3}$A = 1mA i.e., current is 2mA at t = 0 and with time it goes to 1mA.

135. Ans (4)



$\Rightarrow \quad C_{ab} = \frac{2C}{3} = \frac{2\varepsilon_0 A}{3d}$

137. Ans (2)

NCERT XII, Part-I, Pg # 109

	2A	$\longrightarrow$	2B	+	C
t = 0	4 atm		0		0
t = 100s	4 −2x		2x		x

Total pressure = 4 − 2x + 2x + x

5 = 4 + x

x = 1 atm

$k = \frac{2.303}{t}\log\left(\frac{4}{4-2x}\right)$

$k = \frac{2.303}{100}\log\left(\frac{4}{2}\right)$

$k = \frac{2.303 \times 0.3010}{100}$

$k = \frac{0.693}{100}$

$k = 6.93 \times 10^{-3}\ s^{-1}$

138. Ans (3)

$r = k[A]^x[B]^y$

By I, II

A = doubled, B = same, r = doubled

$\therefore x = 1$

By II, III

A = same, B = doubled, r = doubled

$\therefore y = 1$

r = k [A] [B]

By I ; $2 \times 10^{-6} = k \times (1 \times 10^{-2}) \times (2 \times 10^{-2})$

$k = 10^{-2}\ mol^{-1}\ L\ sec^{-1}$ (order = 2)

139. Ans (4)

$$k = \left(\frac{k_1 . k_2}{k_3}\right)^{2/3} ; E = \frac{2}{3}[E_{a_1} + E_{a_2} - E_{a_3}]$$

$$\Rightarrow \frac{2}{3}[180 + 80 - 50] = 140\ \text{kJ/mol}$$

140. Ans (2)

$$\frac{w}{E} = \frac{Q}{96500}$$

142. Ans (3)

Explanation

- Conductivity (κ): It decreases with dilution because the number of ions per unit volume decreases.
- Molar Conductivity (Λm): Defined as Λm = κ / C, it increases with dilution because ion mobility increases due to reduced inter-ionic attractions.
- Strong vs. Weak Electrolytes: Strong electrolytes show a linear increase in Λm with √C, while weak electrolytes follow the Kohlrausch's Law and exhibit a sharp increase due to incomplete dissociation at higher concentrations.

Concept

1. Correct, Conductivity (κ) always decreases with decrease in concentration, for both strong and weak electrolytes because κ α C.
2. Correct, Variation in molar conductivity with concentration, strong electrolytes show a different trend compared to weak electrolytes.
3. Incorrect, Molar conductivity (Λm) increases with dilution, not decreases.
4. Correct, At infinite dilution (C ⟶ 0), molar conductivity is called limiting molar conductivity (Λm°).

Answer Option 3.

143. Ans (2)

For n eq. of substance deposited/converted, nF charge is required.

144. Ans (4)

	$A_3B \longrightarrow$	$3A$ +	B	$x = 0.9$
t_0	1	–	–	$DT_D = \times i \times K_D \times m$
t_0	$1 - x$	$3x$	x	$1 - x + 3x + x = i = 1 + 3x$

146. Ans (1)

NCERT (2019), Pg # 57

150. Ans (4)

NCERT-XII, Part-1, Pg # 135

164. Ans (4)

$CH_2=CH-Cl \longrightarrow$ IUPAC name is 1-Chloro ethene, no other structure is possible for this.

166. Ans (3)

$$CH_3-\overset{*}{C}H(OH)-\overset{*}{C}H(OH)-COOH$$

$2^2 = 4$

174. Ans (1)

Fact

176. Ans (4)